The Seismometer

Pupil sheet

Name: _____ **Date:** _____

Safety: I strike the table, never the box; nothing heavy or thrown. In the room we walk, never run.

1. What I see on the screen

The screen shows **Peak amplitude** (in g), **Shaking index** and a **seismogram** — the last ten seconds, on a fixed ±0.25 g scale. Even in silence the line trembles by about 0.01 g: that is the sensor's own noise.

The shaking index is not a magnitude. Each unit up means shaking ten times stronger, but the index does not know how far away the strike happened — and the distance correction is exactly what makes a magnitude. **It measures how much it shook *here***, something akin to *intensity*, not to magnitude. I use it to compare strikes with each other, not to announce an earthquake.

The silence rule: ten seconds in which **nobody touches the table**. It is announced aloud and counted.

The noise floor (silence, 10 s) = _____ **g** A peak smaller than that *cannot be declared detected.*

2. What I do on my own

First I predict. Will the box feel a footstep taken five metres away? ☐ yes ☐ no
From five metres the amplitude will be about _____ times smaller than from one metre.

because _____

_____

The sources, all from one metre. Three strikes each, with silence between them; after each one I press **Save the peak**.

Source	1	2	3	Mean (g)
Finger, light				
Fist				
Footstep				
Door closed				

The same strike, three distances. I use *the same* source — best of all a book dropped from the same height.

Distance	1	2	3	Mean (g)
1 m				
2 m				
5 m				

How many times it falls from 1 m to 5 m: _____ : _____ = _____ times

I draw the wave shape for the strike at 1 m and for the one at 5 m. What changes — the height, or the duration too?

at 1 m

at 5 m

3. The conclusion I write

I predicted that, and the data show that _____

_____

The shaking index is not a magnitude because _____

_____

Real seismographs are put on bedrock, in basements, because _____

_____

Exit ticket

The same knock gave two different numbers. Why are both correct?

_____